

Software engineer with **12 years in software** and five years building production web products. I cover the delivery path most teams split across three people: the interface, the API layer that feeds it, and the AWS infrastructure it runs on. Current focus is **React, Next.js, and Node.js** systems that stay fast and cheap under real traffic.

10,000+ students served

1 Cr+ BDT annual revenue

2.2M active users reached

0s deploy downtime

TECHNICAL SKILLS

LANGUAGES	JavaScript, TypeScript, Python, Java, Kotlin, Go, PHP
FRONTEND	React, Next.js (App Router), Redux, Tailwind CSS, Sass, React Native
BACKEND	Node.js, Express, FastAPI, REST API design, Socket.io, Prisma, Mongoose, BullMQ
DATA	MongoDB, PostgreSQL, MySQL, Redis, Pinecone (vector search)
CLOUD & OPS	AWS (EC2, S3), GCP Cloud Run, Cloudflare (Workers, CDN, D1), Docker, Nginx, MinIO, blue-green deployment, Linux server administration
RELIABILITY	Sentry, Grafana, Prometheus, Loki, Jest, Git

EXPERIENCE

Learn with Sumit

April 2021 - Present · Remote

Jr. Software Engineer

Ed-tech company running a subscription learning platform for the Bangladeshi developer community.

- Build and maintain a custom **learning management system on Next.js and Node.js** serving **10,000+ students** and carrying more than **1 crore BDT of annual revenue**, working across the frontend, the API layer, and deployment.
- Shipped **DRM-protected HLS video delivery** with watch-time tracking, so course progress and completion are measured from actual playback rather than page visits.
- Replaced password login with **passwordless authentication** using magic links and passkeys, and added single sign-on across platform properties.
- Delivered revenue and retention features end to end: payment gateway integration, a multi-course enrollment system, quiz and assignment flows, a student CV generator, and real-time notifications.
- Secured student and payment data with **JWT authentication, role-based access control**, and encrypted MongoDB storage.
- Deployed and operate the stack on **AWS**, with **Cloudflare Workers** handling certificate generation at the edge to keep that workload off the origin server.

Independent Contract

2025 - Present · Remote

Frontend & Platform Engineer

Offer Finder, the bilingual campaign and merchant discovery product for bKash, Bangladesh's largest mobile financial service.

- Built the **Next.js 16 server-rendered frontend** for a discovery platform that reached approximately **2.2 million active users in Bangladesh** over a one-month window (GA4, 4 August to 2 September 2026).
- Shipped **Bengali-first bilingual routing**: one page tree serves both languages through middleware rewrites and a request-scoped language header, so English and Bengali stay in sync without a duplicated route set.
- Cut page weight by moving filtering to the server. Merchant search stopped shipping **over 15 MB** to the browser, and campaign pages stopped requesting roughly **12 MB** of brand data in a single response, replaced with bounded queries, field selection, request cancellation, and pagination.
- Added **GPS-assisted discovery**: browser coordinates are reverse-geocoded, then matched against backend offer locations to narrow categories and offer types before a filtered campaign URL is generated.
- Designed a **zero-downtime blue-green deployment** on a single EC2 instance with Docker, Nginx, and health checks, removing the 2 to 4 minutes of downtime each rebuild used to cause, at no additional infrastructure cost.
- Hardened delivery for traffic spikes with Cloudflare caching, upstream request timeouts, clustered Node workers, and **Sentry** monitoring across browser, server, and edge runtimes.

EXPERIENCE (CONTINUED)

Analyzen

August 2021 - January 2025 · Dhaka, Bangladesh

Jr. Software Engineer (Aug 2022 - Jan 2025) · Jr. Frontend Developer (Aug 2021 - Oct 2022)

Digital solutions agency delivering web and mobile products for enterprise clients.

- Built production **React and Next.js** interfaces for enterprise clients, translating high-fidelity Figma specifications into accessible components that held up across screen sizes.
- Delivered **Unilever Frontline Academy**, a learning platform for Unilever's frontline workforce across Bangladesh, alongside client projects in legal services and e-commerce.
- Developed interactive dashboards backed by **REST API integration** and non-trivial client-side state, and wrote the Node.js and Express services behind several of them.
- Took over deployment work for the frontend team, shortening release cycles and reducing the manual steps between a merged branch and a live site.
- Promoted from Jr. Frontend Developer to Jr. Software Engineer after 14 months, on delivery record and scope of ownership.

WeblyWork Group

August 2017 - June 2021 · Lalmonirhat, Bangladesh

Android Developer

- Built and released client Android applications in **Java and Kotlin**, covering the full cycle from requirements through Play Store release and post-launch fixes.
- Handled client communication on scope and revisions directly, without a project manager in between.

SELECTED PROJECTS

Distributed HLS Streaming Platform

github.com/saadh393/distributed-hls-streaming-platform

Five-service pipeline that carries a raw upload through queue-based orchestration, FFmpeg transcoding into four HLS renditions with AES-128 encrypted 4-second segments, and adaptive bitrate playback in the browser. Built to locate the failure points of a distributed media pipeline before needing one in production, with the remaining gaps documented rather than hidden.

Stack: Node.js, TypeScript, Express, FFmpeg, BullMQ, Redis, MinIO, PostgreSQL, Docker Compose

Semantic Search API for Amazon Product Data

github.com/saadh393/Semantic-Search

Two-stage retrieval over 12,000 product records: vector similarity returns the top 20 candidates from Pinecone, then an optional Gemini pass reranks them by intent. Cached queries return in roughly 20ms, direct search in 350 to 500ms. Deployed on GCP Cloud Run with a React and TypeScript frontend.

Stack: Python, FastAPI, Pinecone, Google Gemini, React, TypeScript, Tailwind CSS, Cloud Run

SSHM, terminal SSH connection manager

github.com/saadh393/sshm

Go CLI that stores SSH hosts by alias and connects through a fuzzy-searchable terminal UI, so IP addresses and long ssh invocations no longer have to be remembered. Also published **env-example-gen** on npm, which generates .env.example files from an existing environment config.

Stack: Go, Cobra, Bubble Tea, Lip Gloss

WRITING & OPEN SOURCE

Adapter and Port Architecture in Two Real Codebases	saadh393.github.io	The Node.js Event Loop, Explained Properly	Medium
HTTP Caching in Node.js with Undici v7	saadh393.github.io	How Diffie-Hellman Became the Backbone of Encryption	Stackademic
Why AI Coding Agents Cost So Much	saadh393.github.io	Understanding Idempotency in Distributed Systems	Medium
JavaScript Execution Context	saadh393.github.io	Concurrency, Parallelism, and Sequential Execution	Medium

Open source: contributed to [react-bits](#) (issue #278), diagnosing a broken search interface in a widely used React pattern library.

Mentoring: one-to-one React and Next.js instruction, project-based, with code review on every submission. Full archive and interactive case studies at saadh393.github.io.